



ASSIGNMENT 1 – REFLECTIVE REPORT

AR Mount Guide

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Introduction

My final year research, which focuses on how contemporary technology like augmented reality might improve how tourists and regular people explore mountains, is detailed in the reflective report that is provided. This project's primary goal is to improve user experience by creating an affordable augmented reality application that allows users to scan their surroundings and see any mountains that are detected in the direction they are pointing the camera. The following projects overlay AR objects in real-world contexts using the device GPS position, compass direction, camera, and fundamental functionalities of the AR fundamental plugin.

The following report deeply explains all the concepts that I took in order to make this AR project successful. I will also go into great detail about why I chose this project. This project was very difficult to work with since it's something that I have never implemented. And there are a lot of issues and obstacles that I will uncover that I faced during this assignment. There are some aspects that I was unable to accomplish due to limitations of AR. To add to that, I will also discuss areas for future improvement and how I can make this project better in the future.

Despite there being a lot of issues, the project's main goal was fulfilled with a lot of research and guidance from my seniors. The report focuses on all of the ideas I came up with to deal with the problems and how my seniors helped me. Additionally, when I was feeling depressed about my project, my supervisor gave me the right direction and encouragement. Interaction with the supervisor and senior, as well as their guidance, were crucial to the project's successful completion.

In addition, while working on this project, I will consider how my knowledge and abilities have grown. I learned a lot from this project, including time management and other methods to problem-solving. This was an incredible journey from researching and writing a literature review to now finally developing software. The overall goal of the AR Mount Guide project is to create augmented reality software that enables the user to identify mountains within their field of vision and present the virtual AR object in an actual setting.

Self Management

Motivation

One of the main issues is that people usually rely on conventional sources, such tour guides, guided books, or online searches, to learn about certain mountains. These methods are necessary, but they require a lot of human effort, which can be time-consuming, and often the information provided is ambiguous and hard to comprehend. In addition, because I admire nature, it is really challenging for me to identify the mountains when I visit my balcony. I therefore desired an application that would enable people to accurately recognize mountains and obtain additional information about them.

Furthermore, augmented reality technology is my main driving force behind the project. AR has always captivated me since it makes it possible to see virtual objects in real-world settings. For example, well-known apps like Instagram and Snapchat use augmented reality (AR) to allow users to add filters to their faces. I've always been fascinated by these kinds of applications, and I've always wanted to make an augmented reality application. Additionally, I have some expertise utilizing Flutter to create mobile applications. I therefore made the decision to use Flutter to create an augmented reality application without being aware of its limitations.

In general, I picked this project to broaden my knowledge and comprehension of contemporary technologies like augmented reality and investigate how they might be used to address practical issues like using AR overlays to identify mountains. My interest in device orientation and geolocation handling also served as motivation. In the end, I think that this project can improve teaching and learning by offering a fun and interactive way to investigate and comprehend the natural world.

Proper Planning and Time management

Before starting any project, proper planning must be done in order to achieve the desire outcome of the project. Inadequate planning could result in project failure, which is obviously not what I wanted. Since I was fascinated with the technology, such as augmented reality, but this topic is larger and more complex in reality. AR is a modern innovation with a lot of topics and knowledge that needed to be gathered before starting to develop the software. I had to thoroughly investigate and learn more about augmented reality by reading journals, articles, and literature reviews.

Since augmented reality played a significant role in our project, I spent a lot of time simply reading the AR docs to increase my understanding of the technology. But this was really stressful, and I used to shake while working on this assignment. I was growing impatient and wondering why I had selected this project. Thus, there was inadequate time management and planning. I was unable to finish my assignment because of this. I got in touch with my supervisor and senior, and they were able to provide me all the information I needed to get motivated again. They have taught me a number of strategies that will enable me to finish the task correctly.

Development of Gantt Chart

Creating a project plan using a Gantt chart is therefore among the finest recommendations. By making changes to my old Gantt chart, I was able to divide my entire project into manageable stages. I saved a ton of time by splitting up all of my responsibilities between weeks two and ten. I was able to see deadlines, monitor progress, and identify any delays thanks to the Gantt chart.

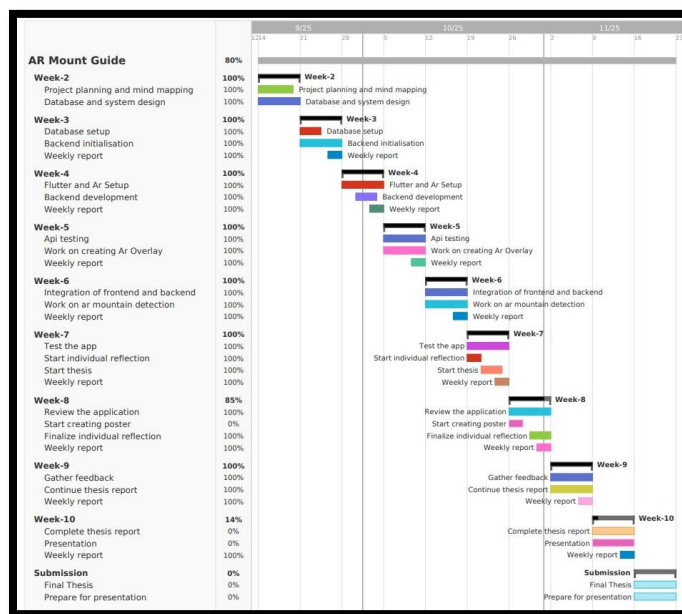


Figure 1: Semester 1 gantt chart

This is my previous Gantt chart, which shows all of the tasks I need to complete between weeks two and eleven. However, this doesn't go into much depth about the task; hence, dissecting the assignment was crucial to effective time management. To ensure that the previous Gantt chart was a comprehensive work structure, I updated and changed it. The redesigned Gantt chart is depicted in the diagram below.

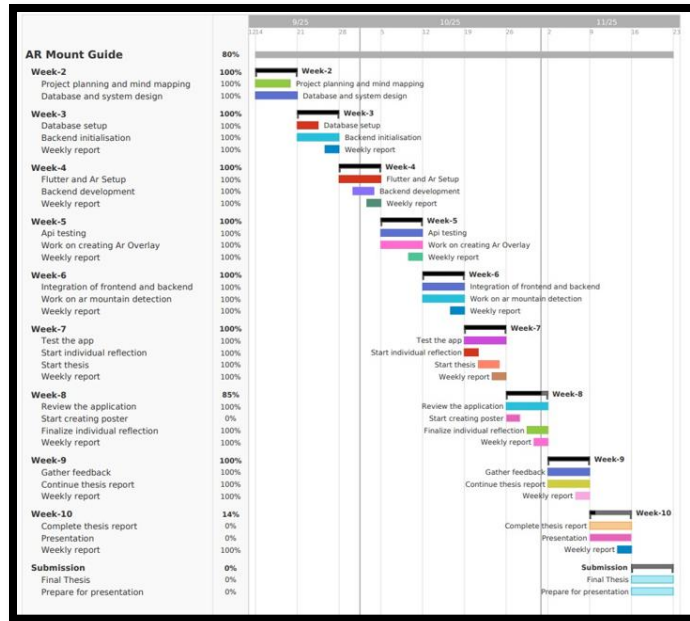


Figure 2: Semester 2 Improvised Gantt chart

Outline Definite Goals and Prioritize Tasks Effectively

Since I mentioned that, I have devoted the majority of my time to learning more about AR applications through study. My progress had been halted as a result, and I was now only in the research stage. My supervisor has thereby improved my mood by giving me advice and encouraging me to "prioritize the task that truly matters". Indeed, putting what really matters first helped to maximize productivity and lower stress levels.

I decided on my main objective and worked toward it step by step. I divided the work and completed each step individually rather than attempting to address the entire issue at once. Additionally, I made careful to refocus on my main objective whenever I felt stuck on a particular subject. I separated my priorities using the MoCoW method. This improved my problem-solving skills and increased my productivity in finishing the project.

Communication and Quality

The most important factor in the AR Mount Guide project's success was good communication with the supervisor. I made sure to stay in constant contact with my supervisor during the research, development, and testing stages. My supervisor, Mr. Krishna Aryal, has supported me in both my good and bad times. He has always been there for me when I needed him and has actively guided

me. He was the one who kept me motivated and guided me with suitable options whenever I encountered problems.

I communicated with my supervisor via WhatsApp and emails. But anytime there was a significant case, we also used Google Meet to communicate. For example, I used Google Meet to connect with my supervisor when the college was closed due to protests in Nepal. Additionally, I updated him with my code file uploaded to GitHub, a version control system. He used to evaluate my GitHub and provide me with the essential input on a regular basis. Furthermore, in the initial phase, there was difficulty in maintaining and managing tasks. So he told me to use Trello since it was easy, cheap, and immediate management for the project. He also reviewed everything and provided me guidance whenever I needed it. He also taught me to prioritize my tasks for an effective development process.

Furthermore, he helped me anytime I made mistakes during my developmental stage. He gave me advice on how to debug and assisted me in solving them. The success of the Ar Mount Guide initiative was largely down to him. Additionally, he gave me a suitable testing plan for this project. Additionally, he helped me with appropriate report structure to ensure consistency and clarity in order to provide high-quality documentation. I double-checked the information in my report with him, and he was really helpful in guiding me. His criticism was the primary factor that kept my report's quality intact.

In conclusion, I was able to make this project successful through efficient communication, the use of a variety of tools, and a methodical approach to preserving quality. In addition, my supervisor's ongoing comments has really aided me in successfully finishing my project.

Reflection

Plan Adjustments and Refinements

I first decided to create a cross-platform software for both iOS and Android. I also had to utilize two separate AR packages because I'm utilizing AR: ARKit for iOS and ARCore for Android. As a result, maintaining two plugins would have been difficult. There was also a hardware limitation for iOS programming since I needed a Mac to build and run on iOS. I decided to focus much more on Android instead of pursuing my goal of developing augmented reality software for iOS. This decision prevented me from having to manage both the ARCore and ARKit plugins. Rather, I just needed to focus on Android AR. However, the issue with the ARCore plugin is not solved yet.

ArCore is effectively managed by Google, but for the past 2 years there has not been a significant update in the ArCore package. Google is not actively maintaining the ARCore plugin, but instead it is maintained by Oleksandr [Leuschenko](#). So it is not actively maintained, and there are several issues that I faced. However, this issue was also solved by looking at various Stack Overflow issues. So `arcore_flutter_plugin` has reported issues due to its deprecated libraries and has a lot of complaints.

Another problem was that, even if it was functioning, initialization crashes occurred frequently. The program used to crash immediately after I used AR, which was a serious problem. It appears that this problem was not yet recognized and was quite challenging to investigate. When I asked my seniors, they cautioned me about the AR problem. `NullPointerException: null cannot be cast to non-null type android.view.View` in `ArCoreView.getView` is the error I got, and it comes from the `arcore` package itself rather than being explicitly programmed. I initially believed that there was a memory leak or something, however this issue was actually being caused by the package itself. Additionally, it usually happens when the AR platform is being disposed of. Since this is a plugin error that hasn't been fixed, I prompted Grok, and the only answer I received was from a guy who used a GitHub fork of the original `ar_core_plugin` to fix the problem. Make a customized, patched version of the plugin to address the particular problems in the project, using your forked Git repository as a guide. Forking enables you to apply patches directly because the original plugin is out of date.

```
arcore_flutter_plugin:  
git:  
  url: https://github.com/mystickminal14/arcore_flutter_plugin.git  
  ref: master # or your branch name
```

Figure 3: Github fork for ar_core_plugin

Writing the following code in Unity and integrating it with Flutter was an additional approach. However, I mainly stick with Flutter because I have less time for my assignment and Unity has a steeper learning curve. However, I would improve and optimize this app in the future.

Furthermore, I previously believed that the Ar Mount Guide Project was entirely dependent on API, so I also constructed it using API. However, there was a problem since it took a bit longer to detect and was not as quick as I had anticipated. For efficient detection, though, I was only working with six mountains. Since just the authentication component is implemented in the backend, I

implemented all of the major functionality in Dart itself. This method made my AR detection quick and easy.

Additionally, at first, I used a 3D model that would be shown when the mountains were identified. It was rendering nicely for a single mountain, but if several mountains were found, the 3D model was rendering slowly. Additionally, the app was running slowly due to excessive battery use. I used a 2D image and placed it inside a 3D sphere because of this. Thus, this method worked well for resolving the problem.

Furthermore, there was one feature, which was indoor simulation, and I had kept this as a feature. But due to my working on both indoor and outdoor. I had to replace this feature with a new feature, which is a normal overlay screen. In the normal overlay screen, the mountain will be detected, but not with AR. It is a normal detector for those not wanting AR.

Development Process and Future enhancement

I divided up my work for the Ar Mount Guide app and worked on each feature in turn. Iterative development accelerated and improved the accuracy of the development process. I started the programming by setting up all the required Flutter permissions for the camera, location, and augmented reality. Following that, I begin working on my project. So I followed agile methodology and managed my task using Trello, a powerful project management tool to manage my task. Trello made my project management capability even stronger and helped me manage my important and less important tasks. Additionally, it managed my progress tracking by determining which tasks are crucial. I did not think of completing this project in one go but instead, I tried to implement it step by step.

Creating a prototype for this application was the first step. I used Figma to create an easy-to-use user interface. After that, I concentrated on setting up AR and began developing an AR application. I first experimented with using AR to create a variety of forms, including spheres, circles, and more. After that, I checked to see if location services were enabled and kept all the permissions in Flutter for an efficient application. After that, I concentrated on tracking device orientation in real-time for AR installation.

In a similar vein, I concentrated on calculating distance by first utilizing the haversine method to determine the bearing between the user's position and each peak. Similar to this, ArContent placement was my primary emphasis, and there were a number of problems because real-time

mountain identification with Ar proved challenging. I discovered a way to recognize mountains with the assistance of a senior and after looking into this problem on Stack Overflow, but it still has some issues when it comes to identifying several mountains. Thus, for the time being, I have selected six mountains. Converting real-world geographic linkages to meaningful AR spatial regions was one of the hurdles. ARCore scene updates are very costly and time-consuming.

I also concentrate on creating my application's backend. However, I eliminated this feature and implemented the essential functionality directly in Flutter itself because geographic and mountain searches were slow. As a result, I developed authentication in Node.js and MongoDB on the advise of my supervisor. Below are the screen shot of the application to demonstrate my current status of the Ar Mount Guide:

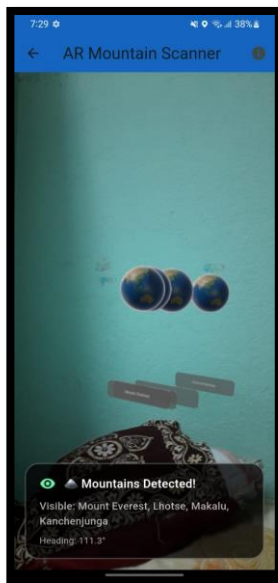


Figure 4: Ar Mountain Overlay

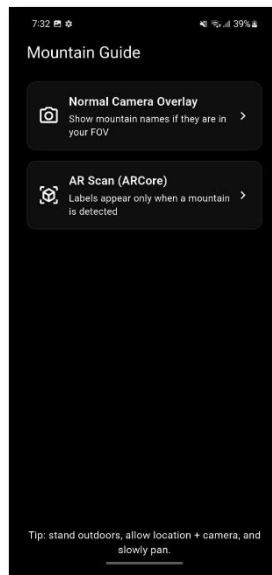


Figure 5: Home screen

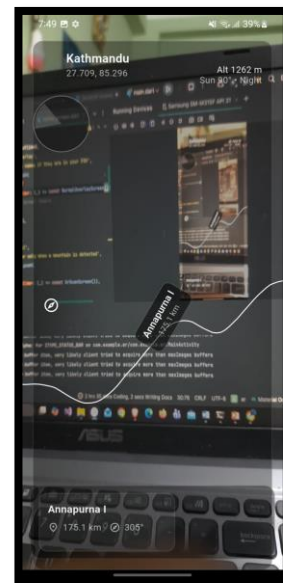


Figure 6: Normal Overlay

By incorporating TensorFlow Lite or PyTorch AMobile to identify actual mountains from camera frames using image classification or a form contour model, we may also use AI in the AR mount guide. Additionally, we could employ both a complete 3D mountain model and a 3D landscape instead of just a standard floating sphere. Additionally, improved 3D rendering and overall performance for the AR application can be achieved if this application is made with Unity. ARCore cannot create AR apps for iOS and Android, but Unity can. To ensure that this study is effective, more mountain data can be collected. Nonetheless, despite having a lot of limitations, the AR Mount Guide app still achieves the required MVP.

Thesis Table of Content

- **Introduction**

Outlines the overall Ar Mount Guide project aims and objectives, problems, and suggested solutions. It also discusses the usage of augmented reality along with the feasibility of the research and challenges that may arise during this project.

- **Literature Review**

It gives a brief overview of the current augmented reality project, emphasizing its importance, utility, and worldwide reach. Additionally, it talks about creating immersive augmented reality experiences for consumers using Flutter.

- **Artifact Planning**

Its primary goal, according to the assigned section, is to analyze requirements using both functional and nonfunctional criteria. It describes the hardware, software, and usability requirements while concentrating on the system design of the program.

- **Development of Artifact and Testing**

It provides thorough guidance on how to create an augmented reality smartphone application, covering everything from setting up the backend to making the AR overlay and including geolocation queries. The various testing strategy and evaluation strategy is also described in depth.

- **Critical Analysis and limitation**

The ultimate conclusion of the Ar Mount Guide project describes the development and research methodology of this thesis. It makes the abilities and information acquired throughout this process further clearer.

- **Future Improvement**

The provided section draws attention to potential future enhancements to this system as well as areas for future feature and marketing additions.

- Conclusion

Overall conclusion of Ar Mount Guide project explaining the journey of this thesis from research to development. It further clarifies the skills and knowleged gained during this development.

Conclusion

To put it briefly, the process of working on this project involved moving from the research stage to the development stage. My ability to conduct research and evaluate other publications, articles, and much more has improved as a result of this undertaking. Additionally, this project has improved my grasp of mobile applications' lifespan and deeper concepts. Along the road, I have also encountered several difficulties and made a few mistakes. However, each error I committed served as a learning opportunity for me.

Additionally, my supervisor and my seniors have been instrumental in this project's successful completion. I was able to finish the task because of their encouragement and feedback. In a similar vein, my endeavor expanded my understanding of augmented reality and its potential to influence the future. As a result, this project was finished with ongoing input from seniors and my supervisor.